Name: Bestha Akshaya Course code: DSA0607

Regno:192424336 Course name: Data handling& visualization

Slot D

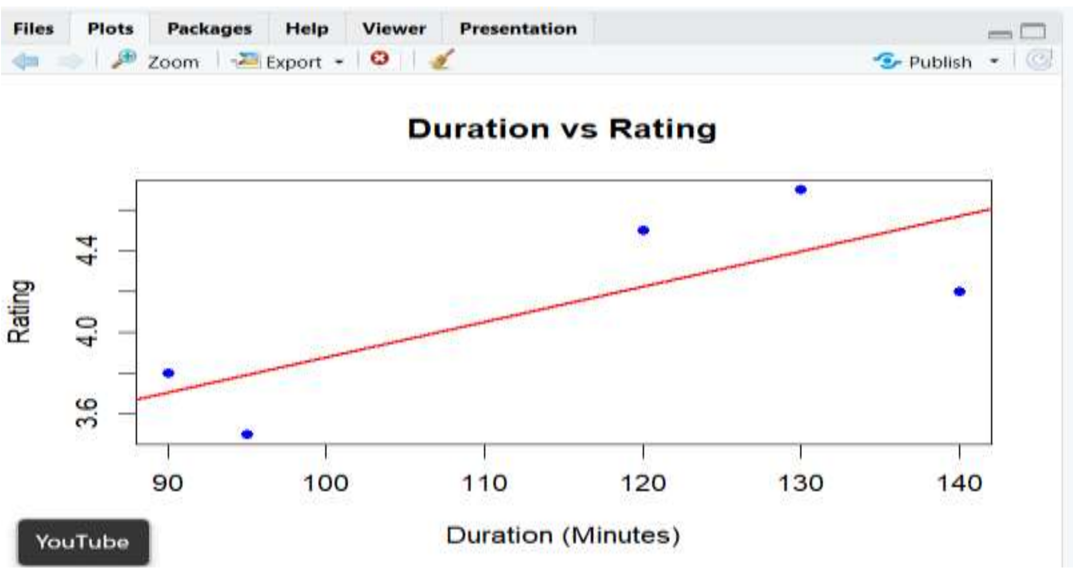
21.

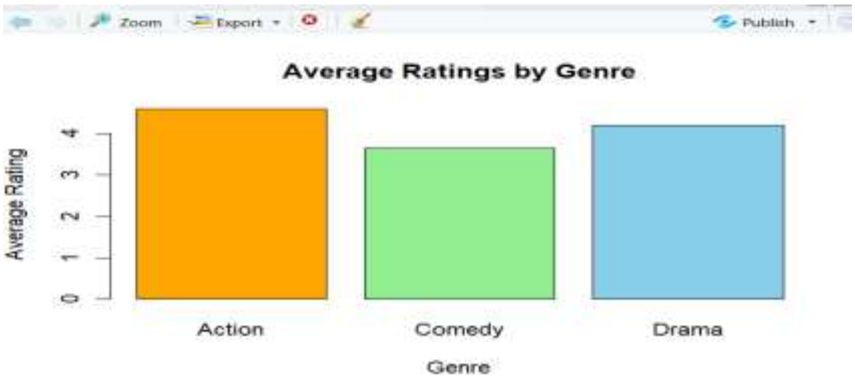
Code:

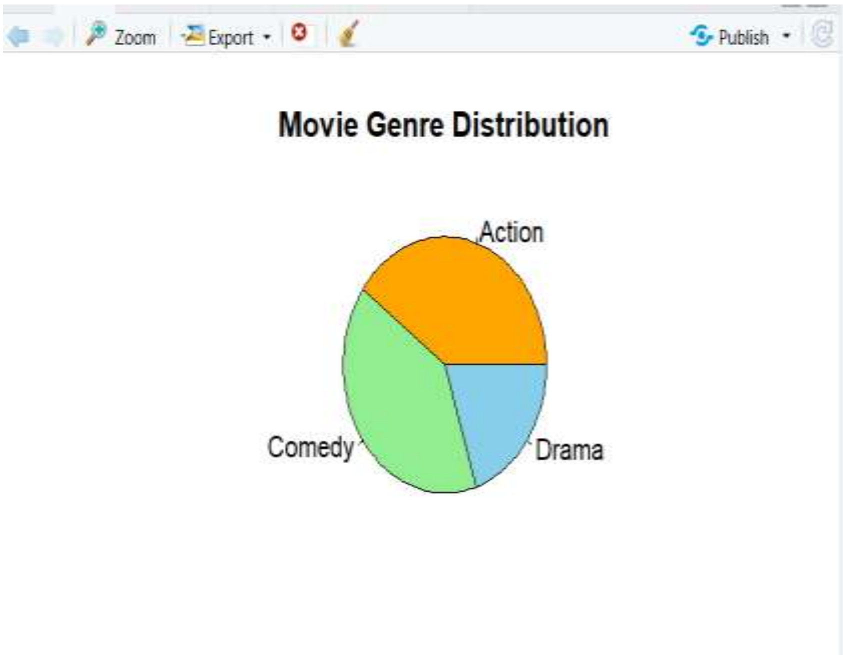
```
Movie_ID <- c(1,2,3,4,5)
Genre <- c("Action",
       "Comedy",
       "Drama",
       "Action",
       "Comedy")
Rating <- c(4.5,3.8,4.2,4.7,3.5)
Duration <- c(120,90,140,130,95)
movie <- data.frame(
  Movie_ID,
  Genre,
  Rating,
  Duration
)
print(movie)
hist(movie$Rating,
     main="Distribution of Movie Ratings",
     xlab="Movie Rating",
     ylab="Frequency",
     col="skyblue",
     border="black")
genre_count <- table(movie$Genre)
pie(genre_count,
    main="Movie Genre Distribution",
    labels=names(genre_count),
    col=c("orange","lightgreen","skyblue"))
```

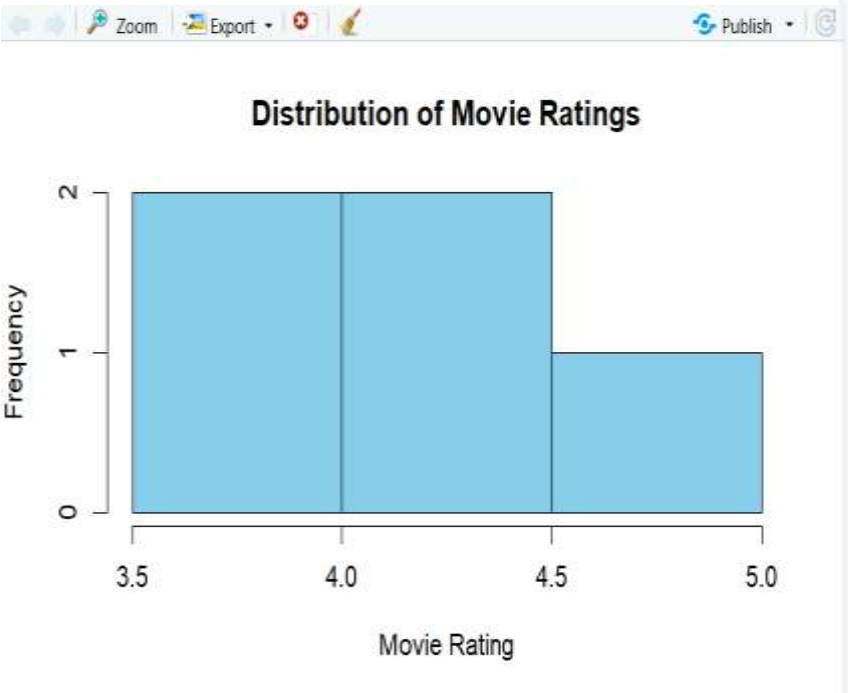
```
avg_rating <- aggregate(Rating ~ Genre,
                        data=movie,
                        mean)
barplot(avg_rating$Rating,
        names.arg=avg_rating$Genre,
        main="Average Ratings by Genre",
        xlab="Genre",
        ylab="Average Rating",
        col=c("orange","lightgreen","skyblue"),
        border="black")
plot(movie$Duration,
     movie$Rating,
     main="Duration vs Rating",
     xlab="Duration (Minutes)",
     ylab="Rating",
     pch=19,
     col="blue")
abline(lm(Rating ~ Duration,
          data=movie),
       col="red",
       lwd=2)
```

OUTPUT:




Zoom
Export
Publish
Average Ratings by Genre
Average Rating
0
1
2
3
4
Action
Comedy
Drama
Genre


Zoom
Export
Publish
Movie Genre Distribution
Action
Comedy
Drama


Zoom
Export
Publish
Distribution of Movie Ratings
Frequency
0
1
2
3.5
4.0
4.5
5.0
Movie Rating

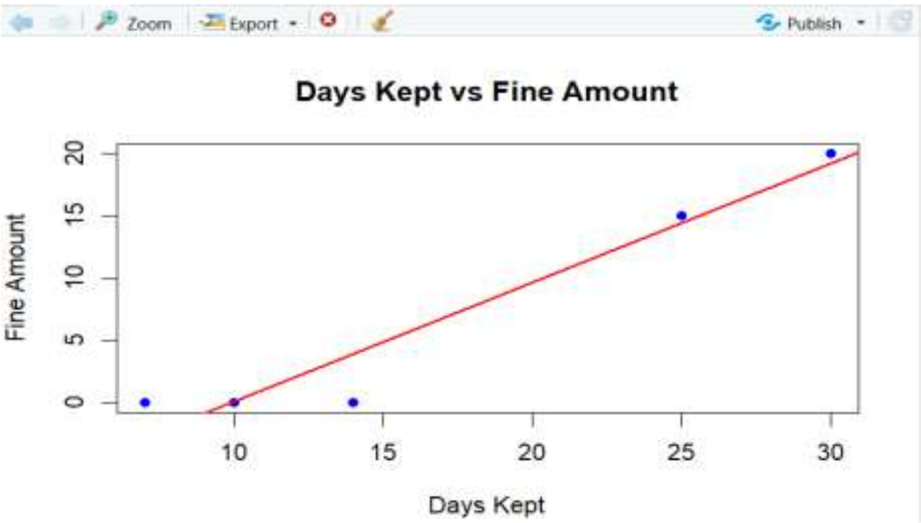
22.

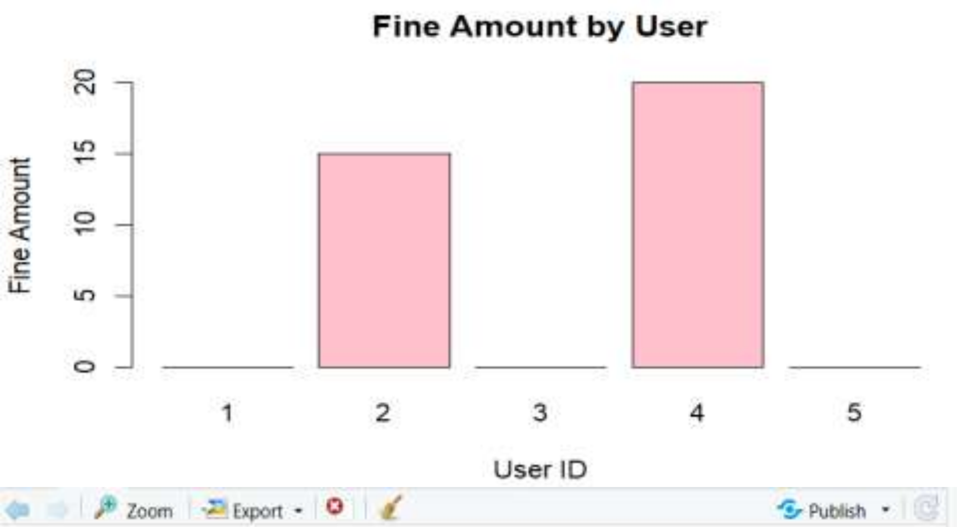
Code:

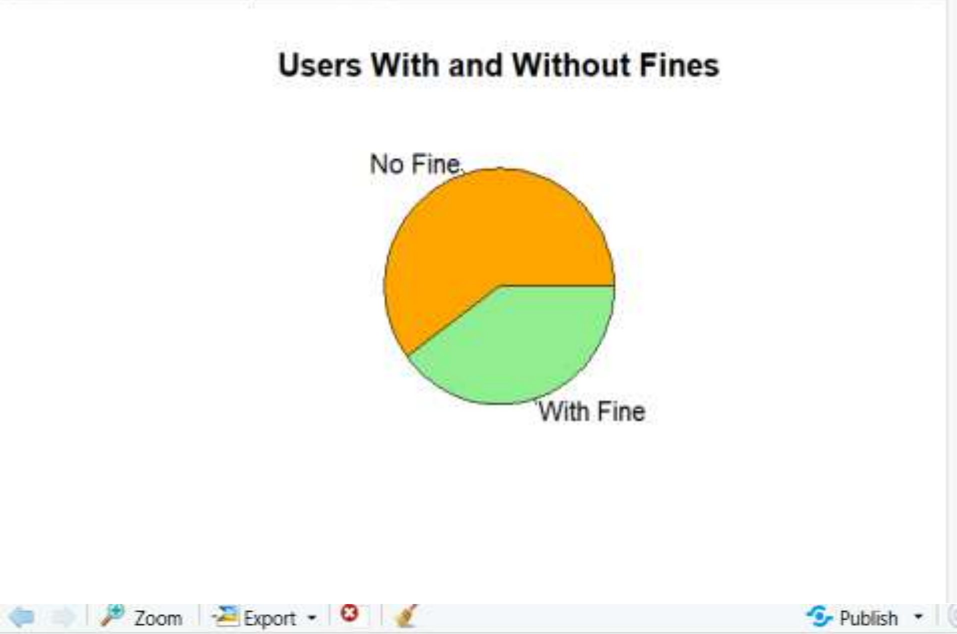
```
User_ID <- c(1,2,3,4,5)
Books_Borrowed <- c(2,5,3,6,1)
Days_Kept <- c(10,25,14,30,7)
Fine_Amount <- c(0,15,0,20,0)
library_data <- data.frame(
  User_ID,
  Books_Borrowed,
  Days_Kept,
  Fine_Amount
)
print(library_data)
hist(library_data$Books_Borrowed,
     main="Distribution of Books Borrowed",
     xlab="Books Borrowed",
     ylab="Frequency",
     col="skyblue",
     border="black")
Fine_Status <- ifelse(library_data$Fine_Amount > 0,
                      "With Fine",
                      "No Fine")

fine_table <- table(Fine_Status)


pie(fine_table,
    main="Users With and Without Fines",
    labels=names(fine_table),
    col=c("orange","lightgreen"))
barplot(library_data$Fine_Amount,
        names.arg=library_data$User_ID,
        main="Fine Amount by User",
```

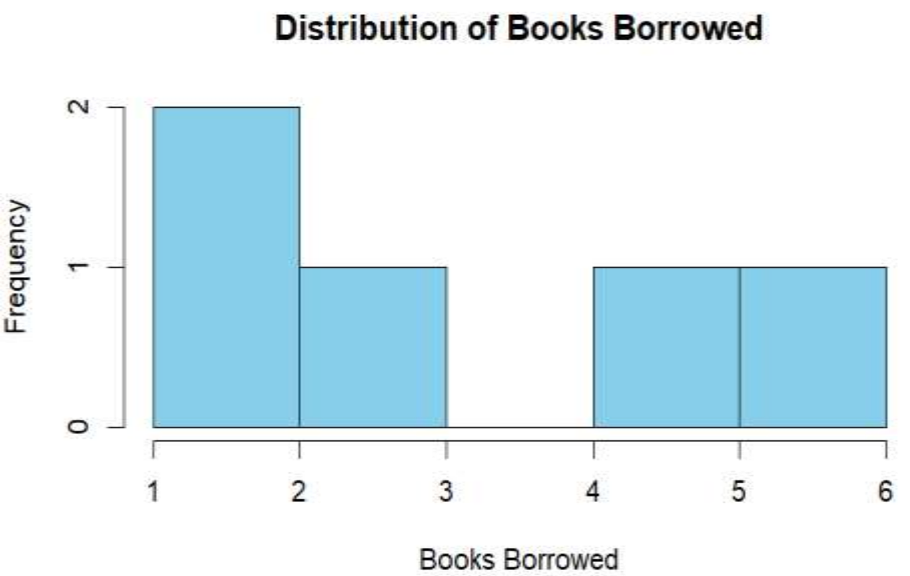
```
        xlab="User ID",
        ylab="Fine Amount",
        col="pink",
        border="black")
plot(library_data$Days_Kept,
     library_data$Fine_Amount,
     main="Days Kept vs Fine Amount",
     xlab="Days Kept",
     ylab="Fine Amount",
     pch=19,
     col="blue")
abline(lm(Fine_Amount ~ Days_Kept,
          data=library_data),
       col="red",
       lwd=2)
```

OUTPUT:




Fine Amount by User
Fine Amount
0
5
10
15
20
1
2
3
4
5
User ID
Zoom
Export
Publish


Users With and Without Fines
No Fine
With Fine
Zoom
Export
Publish


Distribution of Books Borrowed
Frequency
0
1
2
1
2
3
4
5
6
Books Borrowed

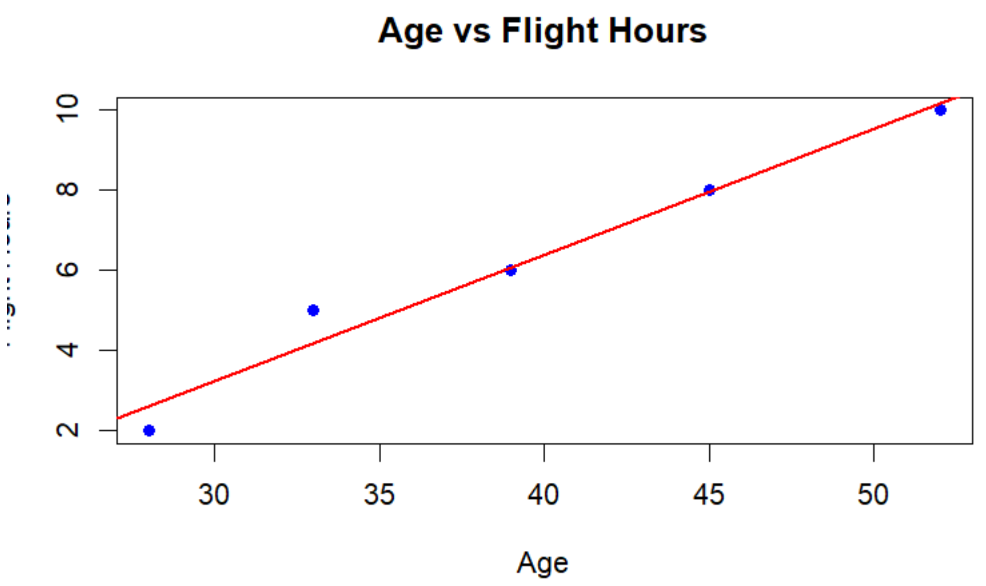
23.

Code:

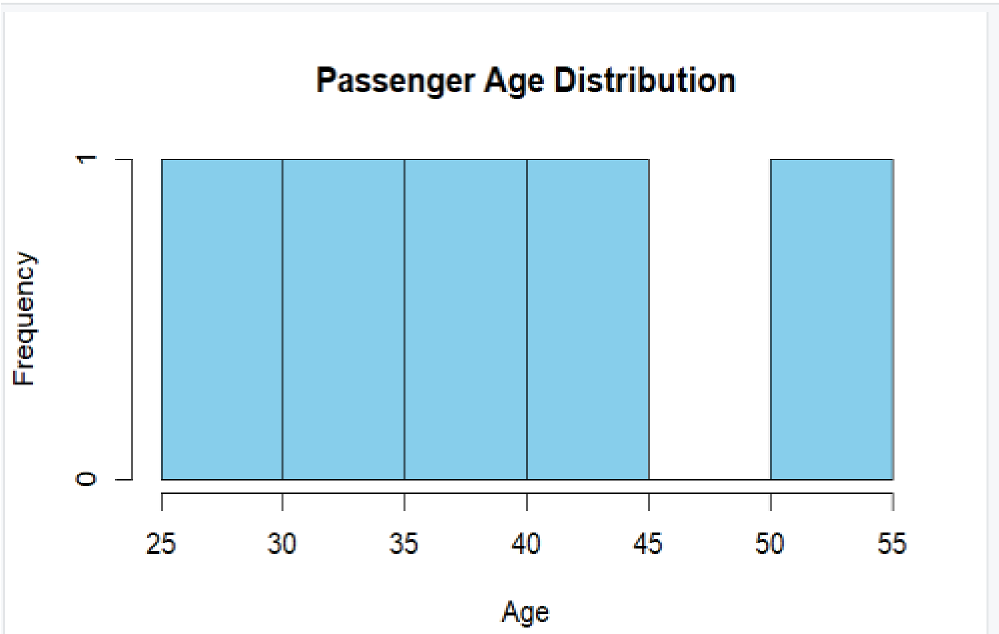
```
Passenger_ID <- c(1,2,3,4,5)
Age <- c(28,45,33,52,39)
Flight_Hours <- c(2,8,5,10,6)
Satisfaction <- c("High",
                  "Medium",
                  "High",
                  "Low",
                  "Medium")
airline <- data.frame(
  Passenger_ID,
  Age,
  Flight_Hours,
  Satisfaction
)
print(airline)
hist(airline$Age,
     main="Passenger Age Distribution",
     xlab="Age",
     ylab="Frequency",
     col="skyblue",
     border="black")
satisfaction_table <- table(airline$Satisfaction)
pie(satisfaction_table,
    main="Passenger Satisfaction Levels",
    labels=names(satisfaction_table),
    col=c("lightgreen","orange","skyblue"))
barplot(airline$Flight_Hours,
        names.arg=airline$Passenger_ID,
        main="Flight Hours by Passenger",
```

```
        xlab="Passenger ID",
        ylab="Flight Hours",
        col="pink",
        border="black")
plot(airline$Age,
     airline$Flight_Hours,
     main="Age vs Flight Hours",
     xlab="Age",
     ylab="Flight Hours",
     pch=19,
     col="blue")
abline(lm(Flight_Hours ~ Age,
          data=airline),
       col="red",
       lwd=2)
```

OUTPUT:



Zoom
Export
Publish
Flight Hours by Passenger
Flight Hours
0
2
4
6
8
10
1
2
3
4
5
Passenger ID
Passenger Satisfaction Levels
High
Low
Medium



Passenger Age Distribution
Frequency
0
1
25
30
35
40
45
50
55
Age

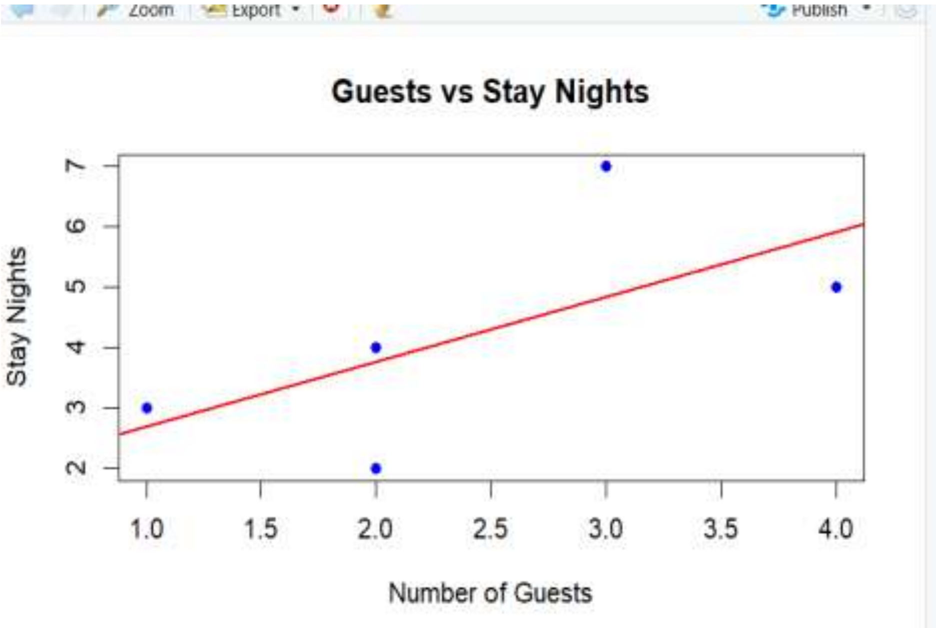
24.

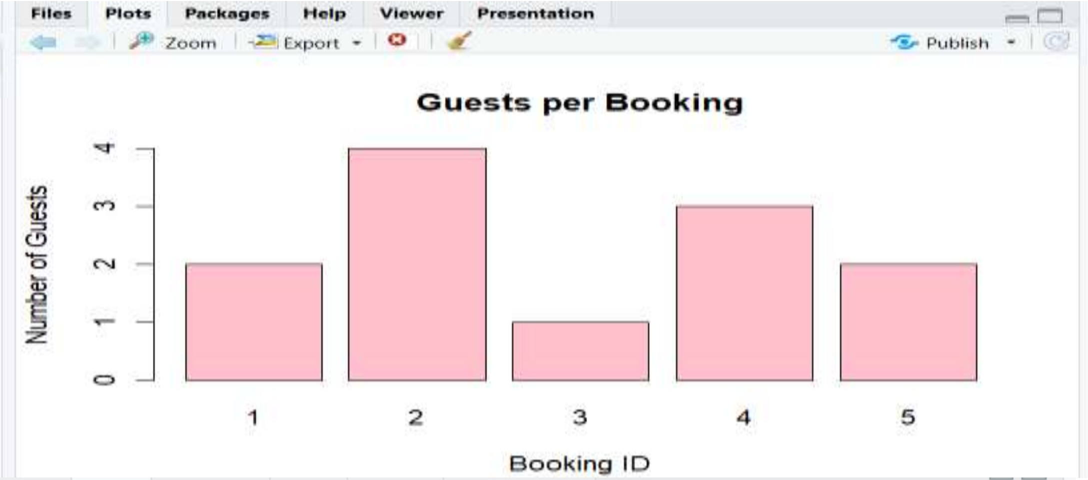
Code:

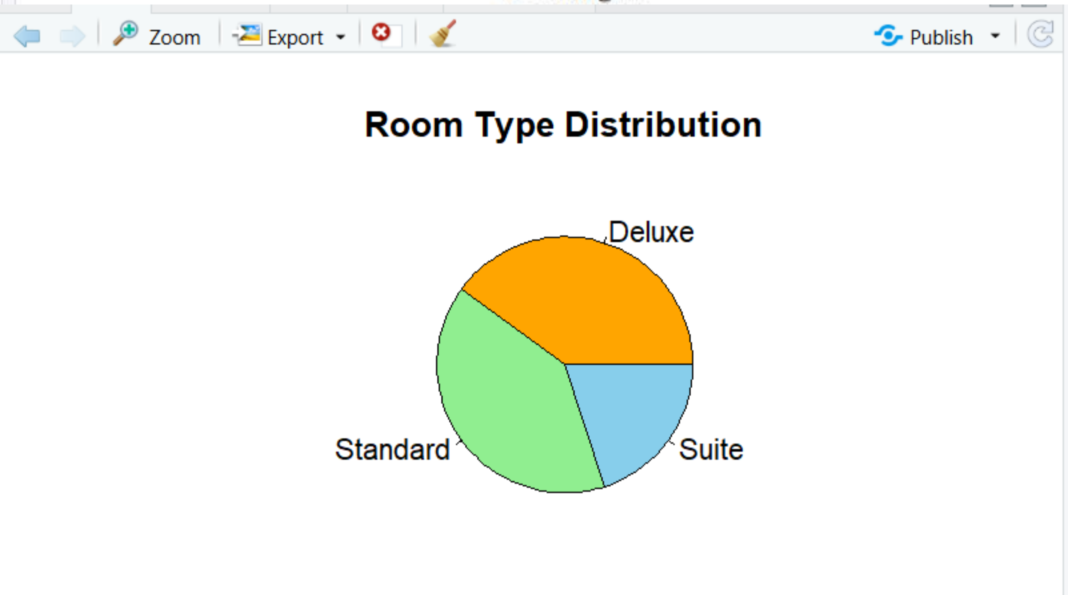
```
Booking_ID <- c(1,2,3,4,5)
Stay_Nights <- c(2,5,3,7,4)
Guests <- c(2,4,1,3,2)
Room_Type <- c("Standard",
               "Deluxe",
               "Standard",
               "Suite",
               "Deluxe")
hotel <- data.frame(
  Booking_ID,
  Stay_Nights,
  Guests,
  Room_Type
)
print(hotel)
hist(hotel$Stay_Nights,
     main="Distribution of Stay Nights",
     xlab="Stay Nights",
     ylab="Frequency",
     col="skyblue",
     border="black")
room_table <- table(hotel$Room_Type)
pie(room_table,
    main="Room Type Distribution",
    labels=names(room_table),
    col=c("orange","lightgreen","skyblue"))
barplot(hotel$Guests,
        names.arg=hotel$Booking_ID,
        main="Guests per Booking",
```

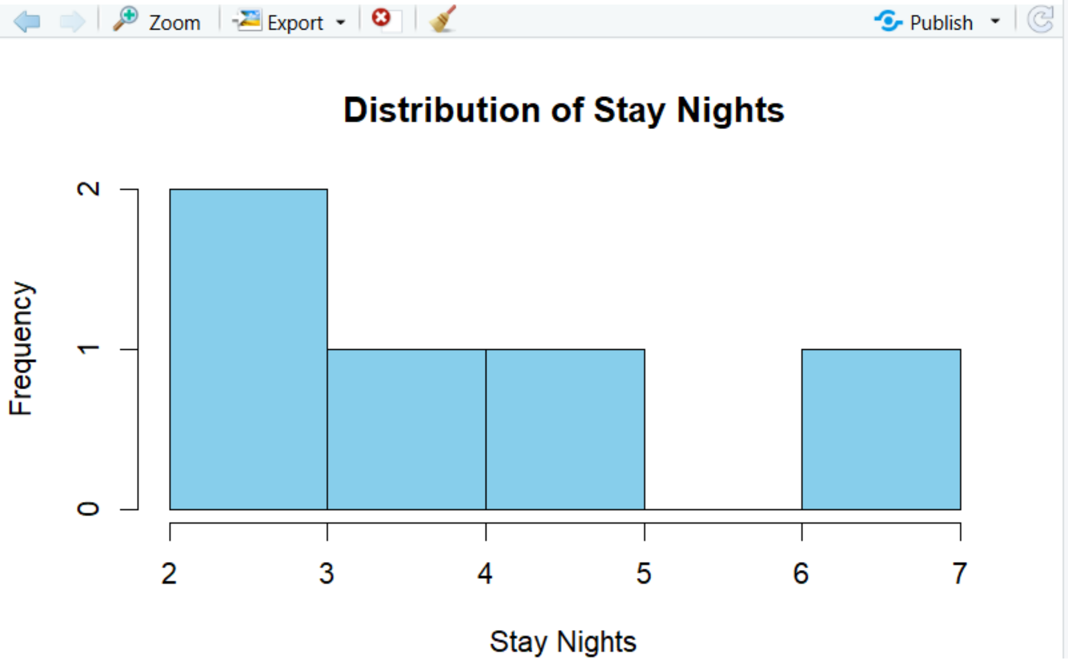
```
        xlab="Booking ID",
        ylab="Number of Guests",
        col="pink",
        border="black")
plot(hotel$Guests,
     hotel$Stay_Nights,
     main="Guests vs Stay Nights",
     xlab="Number of Guests",
     ylab="Stay Nights",
     pch=19,
     col="blue")
abline(lm(Stay_Nights ~ Guests,
          data=hotel),
       col="red",
       lwd=2)
```

OUTPUT:




Files
Plots
Packages
Help
Viewer
Presentation
Zoom
Export
Publish
Guests per Booking
Number of Guests
0
1
2
3
4
1
2
3
4
5
Booking ID


Zoom
Export
Publish
Room Type Distribution
Deluxe
Standard
Suite


Zoom
Export
Publish
Distribution of Stay Nights
Frequency
0
1
2
2
3
4
5
6
7
Stay Nights

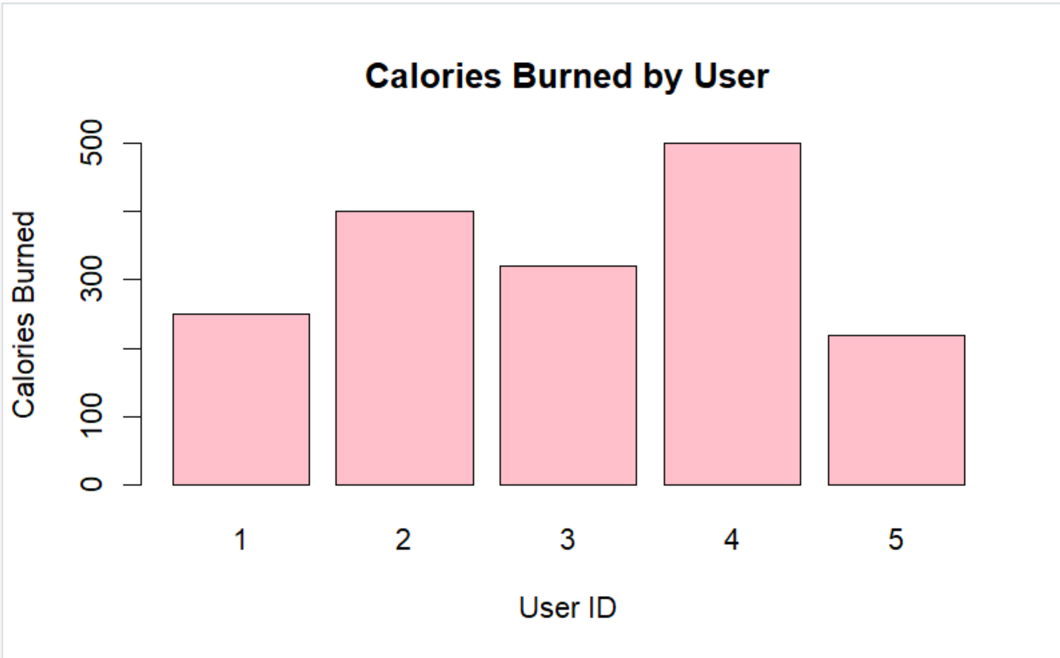
25.

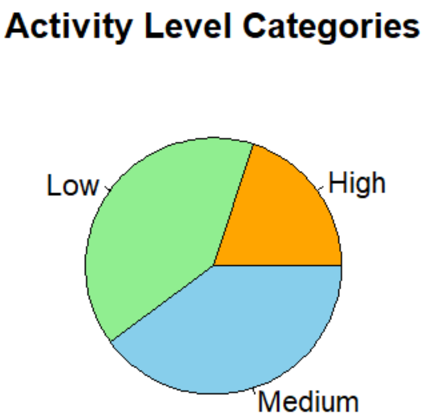
Code:

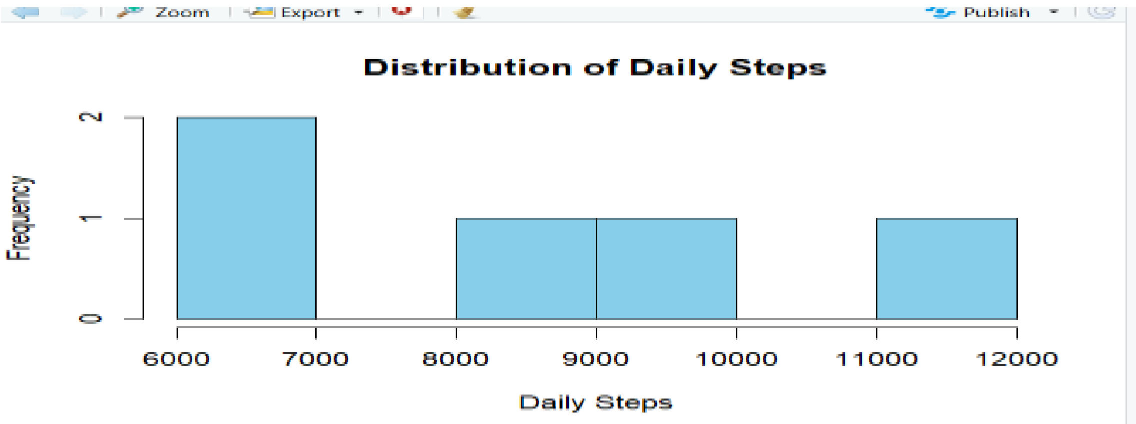
```
User_ID <- c(1,2,3,4,5)
Steps <- c(7000,10000,8500,12000,6500)
Calories_Burned <- c(250,400,320,500,220)
Active_Minutes <- c(40,60,50,75,35)
fitness <- data.frame(
  User_ID,
  Steps,
  Calories_Burned,
  Active_Minutes
)
print(fitness)
hist(fitness$Steps,
     main="Distribution of Daily Steps",
     xlab="Daily Steps",
     ylab="Frequency",
     col="skyblue",
     border="black")
Activity_Level <- ifelse(fitness$Active_Minutes < 45,
                         "Low",
                         ifelse(fitness$Active_Minutes <= 60,
                                "Medium",
                                "High"))

activity_table <- table(Activity_Level)

pie(activity_table,
    main="Activity Level Categories",
    labels=names(activity_table),
    col=c("orange","lightgreen","skyblue"))
barplot(fitness$Calories_Burned,
```

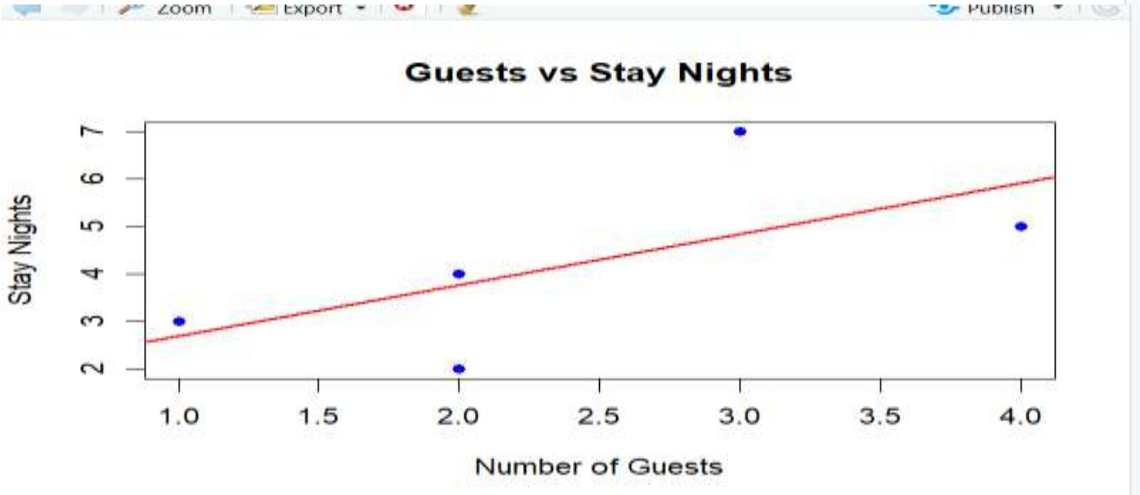
```
names.arg=fitness$User_ID,
main="Calories Burned by User",
xlab="User ID",
ylab="Calories Burned",
col="pink",
border="black")
```

OUTPUT:









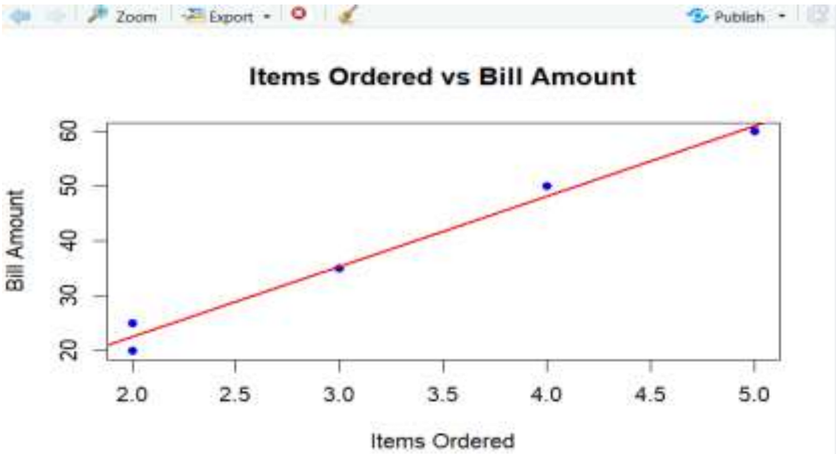
26.

Code:

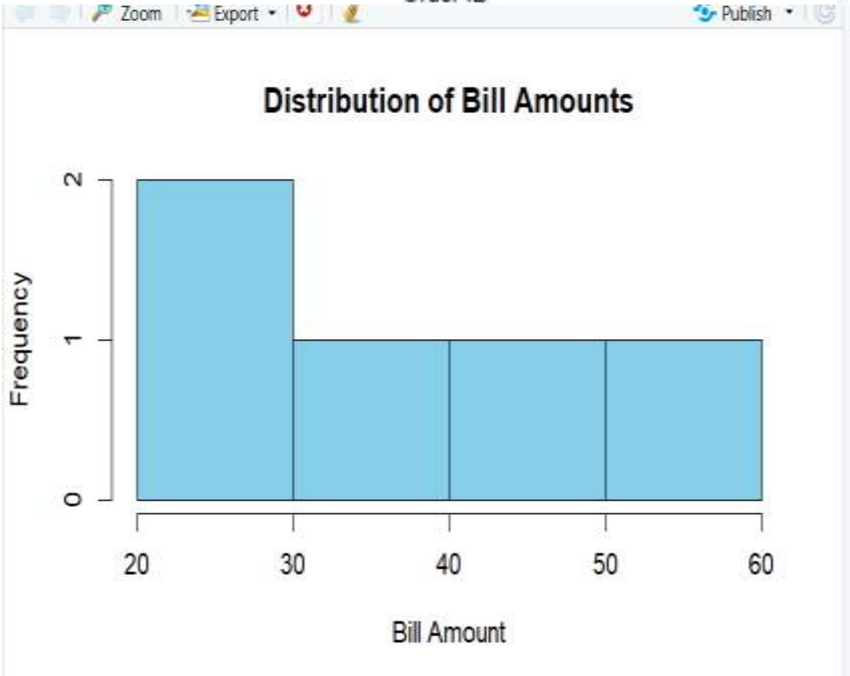
```
Order_ID <- c(1,2,3,4,5)
Items_Ordered <- c(2,5,3,4,2)
Bill_Amount <- c(25,60,35,50,20)
Dining_Type <- c("Dine-In",
                 "Takeaway",
                 "Dine-In",
                 "Delivery",
                 "Takeaway")
restaurant <- data.frame(
  Order_ID,
  Items_Ordered,
  Bill_Amount,
  Dining_Type
)
```

```
print(restaurant)
hist(restaurant$Bill_Amount,
     main="Distribution of Bill Amounts",
     xlab="Bill Amount",
     ylab="Frequency",
     col="skyblue",
     border="black")
dining_table <- table(restaurant$Dining_Type)

pie(dining_table,
    main="Dining Type Distribution",
    labels=names(dining_table),
    col=c("orange","lightgreen","skyblue"))
barplot(restaurant$Items_Ordered,
        names.arg=restaurant$Order_ID,
        main="Items Ordered by Order",
        xlab="Order ID",
        ylab="Items Ordered",
        col="pink",
        border="black")

plot(restaurant$Items_Ordered,
     restaurant$Bill_Amount,
     main="Items Ordered vs Bill Amount",
     xlab="Items Ordered",
     ylab="Bill Amount",
     pch=19,
     col="blue")
abline(lm(Bill_Amount ~ Items_Ordered,
          data=restaurant),
       col="red",
       lwd=2)
```

OUTPUT:


Zoom
Export
Publish
Items Ordered vs Bill Amount
Bill Amount
20 30 40 50 60
2.0 2.5 3.0 3.5 4.0 4.5 5.0
Items Ordered


Zoom
Export
Publish
Items Ordered by Order
Items Ordered
0 1 2 3 4 5
1 2 3 4 5
Order ID


Zoom
Export
Publish
Distribution of Bill Amounts
Frequency
0 1 2
20 30 40 50 60
Bill Amount

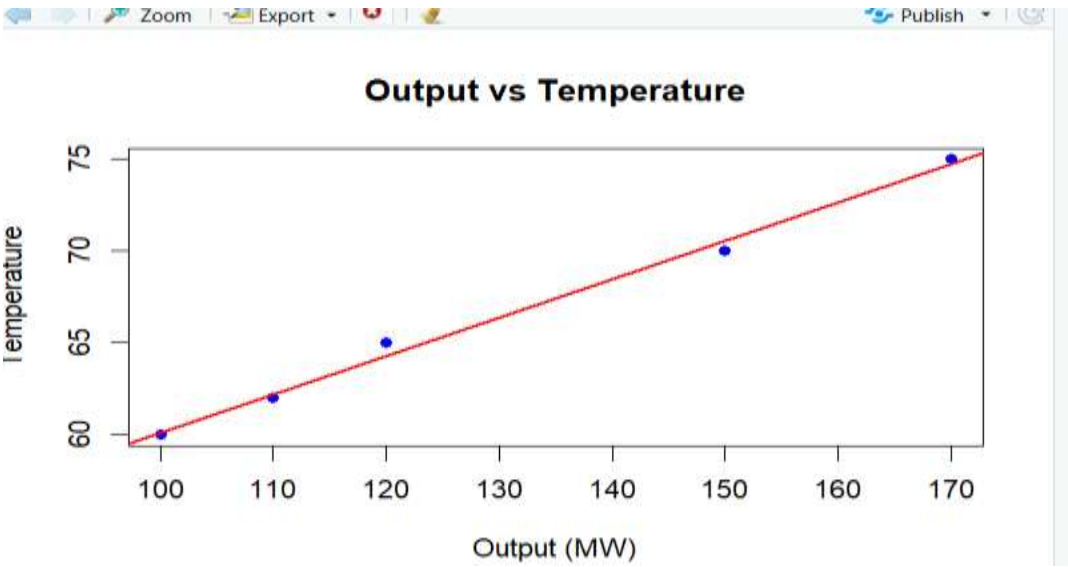
27.

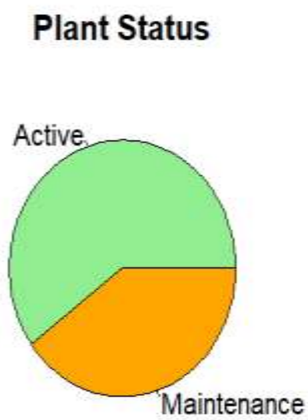
Code:

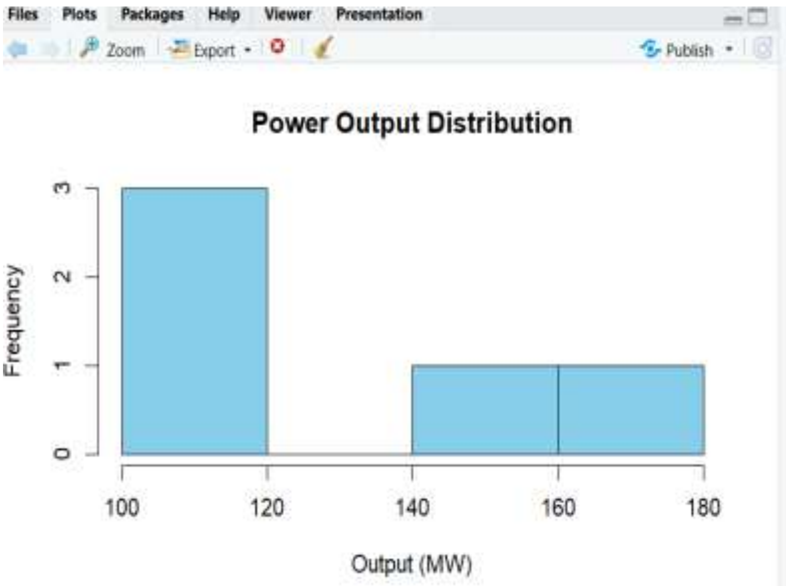
```
Plant_ID <- c(1,2,3,4,5)
Output_MW <- c(120,150,100,170,110)
Temperature <- c(65,70,60,75,62)
Status <- c("Active","Active","Maintenance","Active","Maintenance")
plant <- data.frame(
  Plant_ID,
  Output_MW,
  Temperature,
  Status
)
print(plant)
hist(plant$Output_MW,
     main="Power Output Distribution",
     xlab="Output (MW)",
     ylab="Frequency",
     col="skyblue",
     border="black")
status_table <- table(plant$Status)

pie(status_table,
```

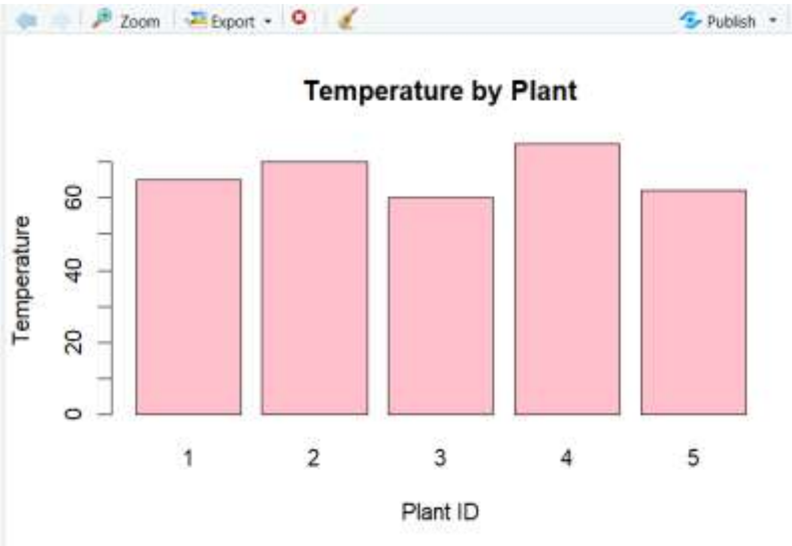
```
    main="Plant Status",
    labels=names(status_table),
    col=c("lightgreen","orange"))
barplot(plant$Temperature,
        names.arg=plant$Plant_ID,
        main="Temperature by Plant",
        xlab="Plant ID",
        ylab="Temperature",
        col="pink",
        border="black")
plot(plant$Output_MW,
     plant$Temperature,
     main="Output vs Temperature",
     xlab="Output (MW)",
     ylab="Temperature",
     pch=19,
     col="blue")
abline(lm(Temperature ~ Output_MW,
          data=plant),
       col="red",
       lwd=2)
```

OUTPUT:




Plant Status
Active
Maintenance


Files
Plots
Packages
Help
Viewer
Presentation
Zoom
Export
Publish
Power Output Distribution
Frequency
0
1
2
3
100
120
140
160
180
Output (MW)


Zoom
Export
Publish
Temperature by Plant
Temperature
0
20
40
60
1
2
3
4
5
Plant ID

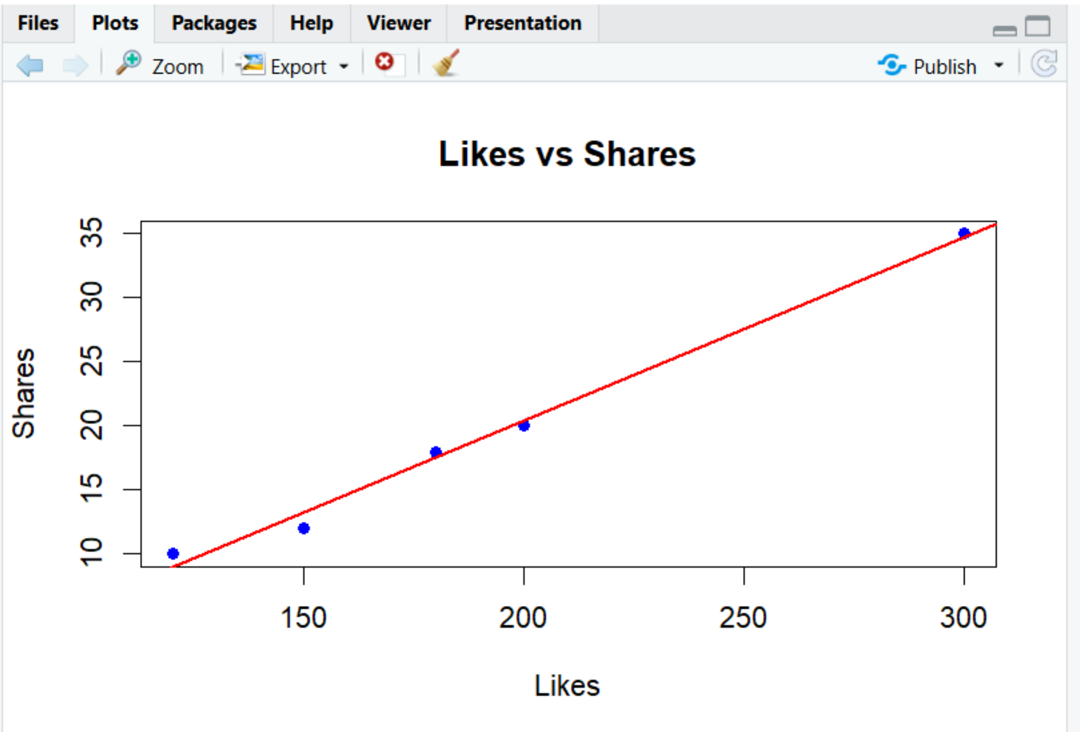
28.

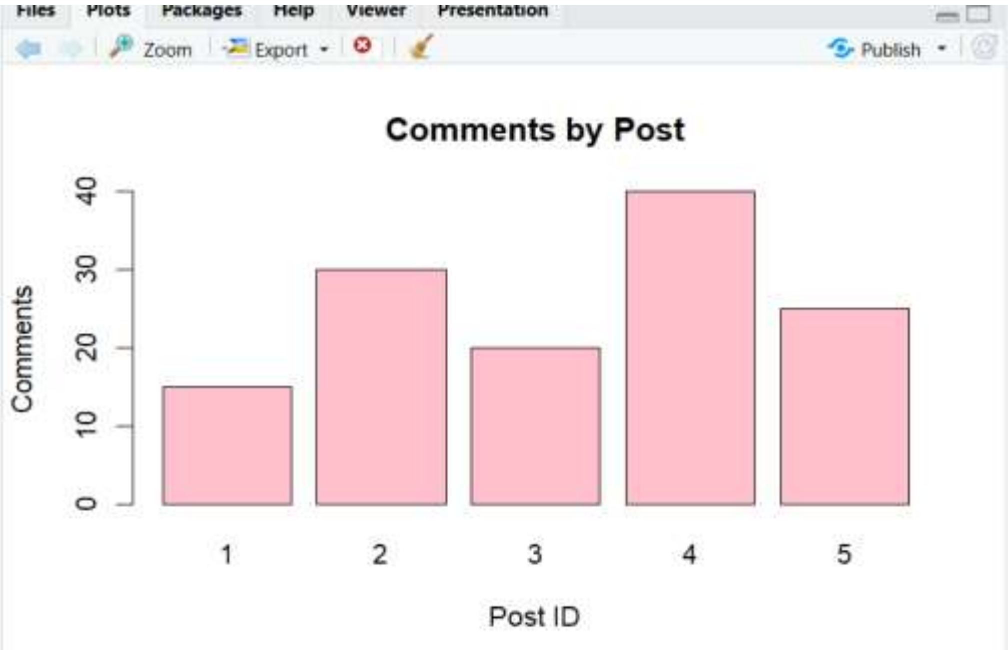
Code:

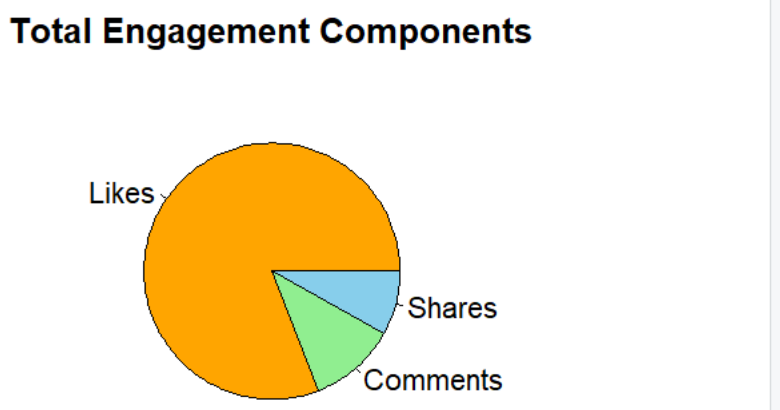
```
Post_ID <- c(1,2,3,4,5)
Likes <- c(120,200,150,300,180)
Comments <- c(15,30,20,40,25)
Shares <- c(10,20,12,35,18)
social <- data.frame(
  Post_ID,
  Likes,
  Comments,
  Shares
)
print(social)
hist(social$Likes,
     main="Distribution of Likes",
     xlab="Likes",
     ylab="Frequency",
     col="skyblue",
     border="black")
engagement <- c(sum(social$Likes),
                sum(social$Comments),
                sum(social$Shares))

names(engagement) <- c("Likes","Comments","Shares")


pie(engagement,
    main="Total Engagement Components",
    labels=names(engagement),
    col=c("orange","lightgreen","skyblue"))
barplot(social$Comments,
        names.arg=social$Post_ID,
```

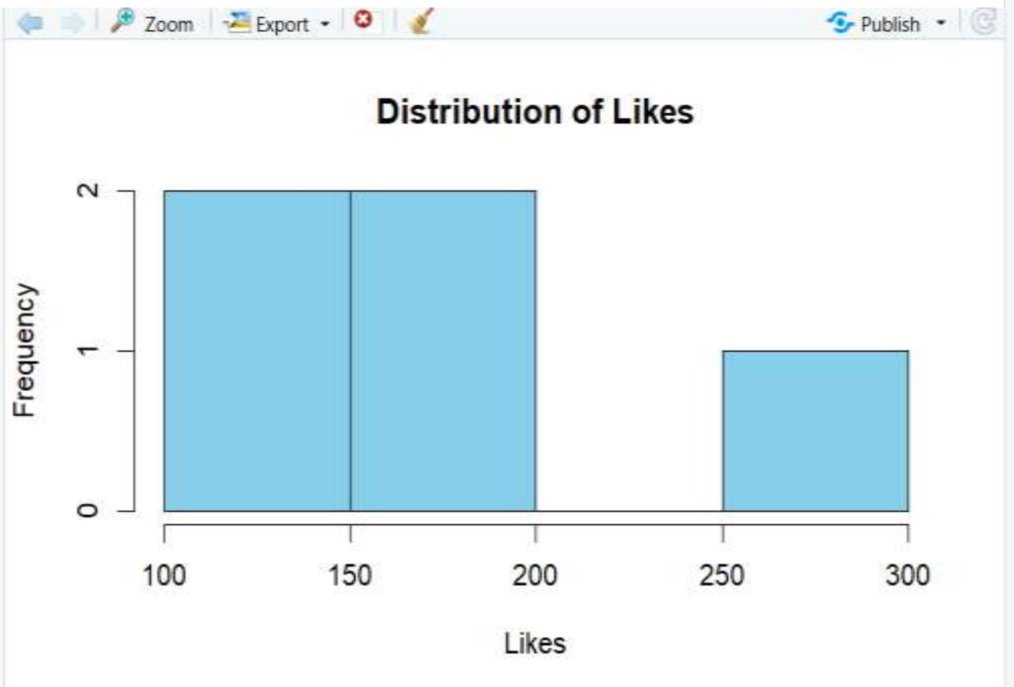
```
        main="Comments by Post",
        xlab="Post ID",
        ylab="Comments",
        col="pink",
        border="black")
plot(social$Likes,
     social$Shares,
     main="Likes vs Shares",
     xlab="Likes",
     ylab="Shares",
     pch=19,
     col="blue")
abline(lm(Shares ~ Likes,
          data=social),
       col="red",
       lwd=2)
```

OUTPUT:




Files Plots Packages Help Viewer Presentation
Zoom Export Publish
Comments by Post
Comments
0 10 20 30 40
1 2 3 4 5
Post ID


Total Engagement Components
Likes
Shares
Comments


Zoom Export Publish
Distribution of Likes
Frequency
0 1 2
100 150 200 250 300
Likes

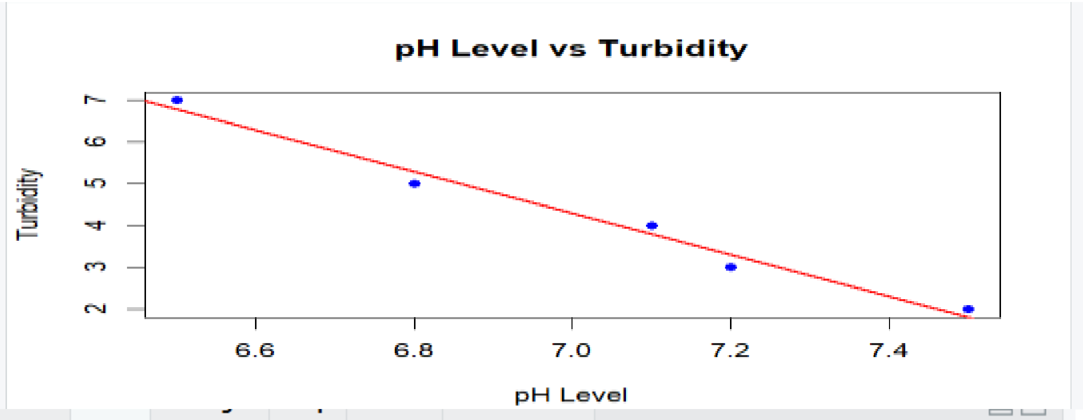
29.

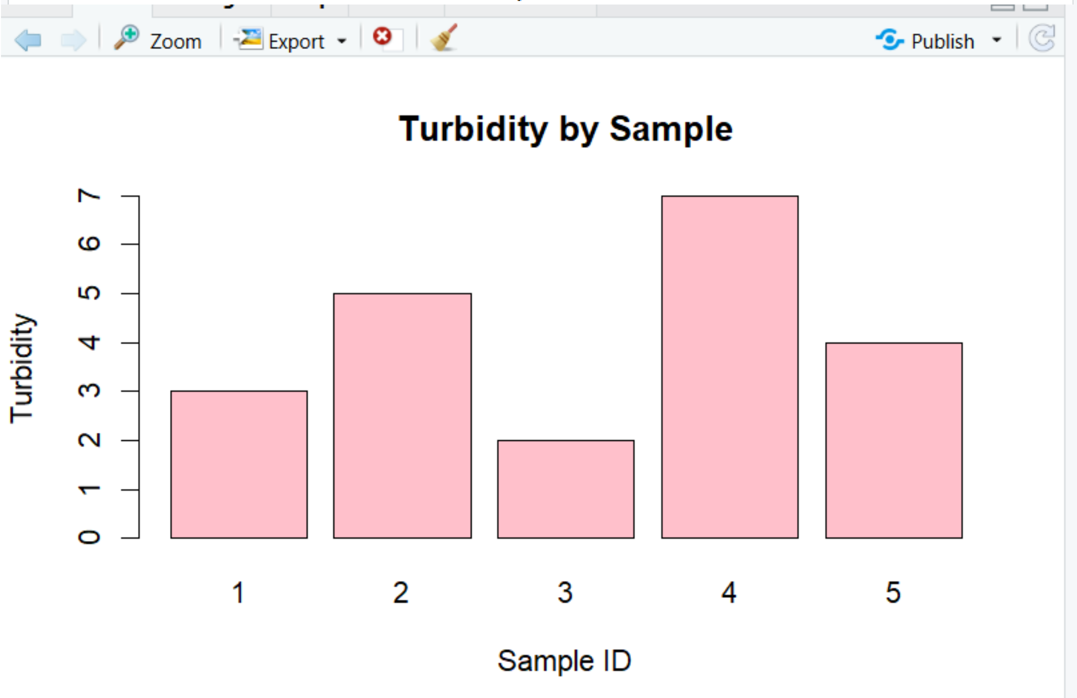
Code:

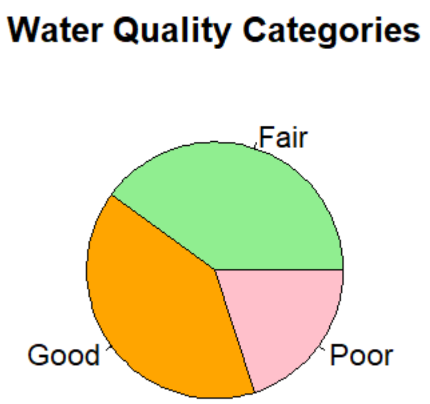
```
Sample_ID <- c(1,2,3,4,5)
pH_Level <- c(7.2,6.8,7.5,6.5,7.1)
Turbidity <- c(3,5,2,7,4)
Quality <- c("Good","Fair","Good","Poor","Fair")
water <- data.frame(
  Sample_ID,
  pH_Level,
  Turbidity,
  Quality
)
print(water)
hist(water$pH_Level,
     main="Distribution of pH Levels",
     xlab="pH Level",
     ylab="Frequency",
     col="skyblue",
     border="black")
quality_table <- table(water$Quality)

pie(quality_table,
    main="Water Quality Categories",
    labels=names(quality_table),
    col=c("lightgreen","orange","pink"))
barplot(water$Turbidity,
        names.arg=water$Sample_ID,
        main="Turbidity by Sample",
        xlab="Sample ID",
        ylab="Turbidity",
        col="pink",
```

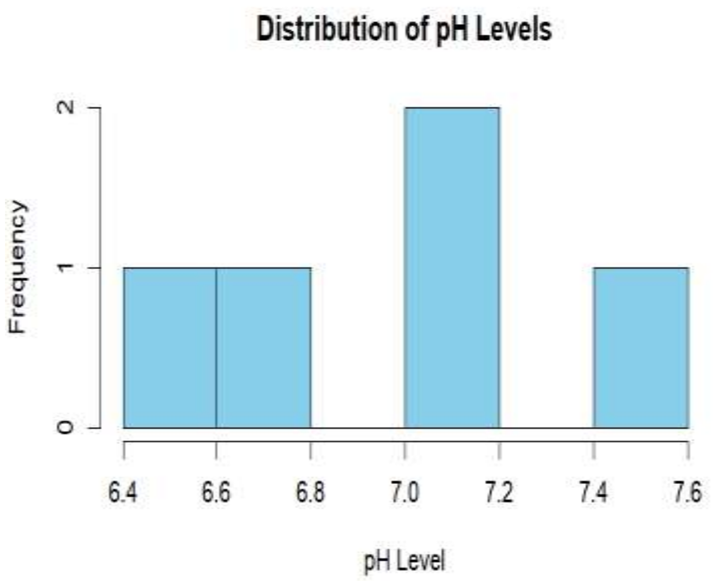
```
          border="black")
plot(water$pH_Level,
     water$Turbidity,
     main="pH Level vs Turbidity",
     xlab="pH Level",
     ylab="Turbidity",
     pch=19,
     col="blue")
abline(lm(Turbidity ~ pH_Level,
          data=water),
       col="red",
       lwd=2)
```

OUTPUT:





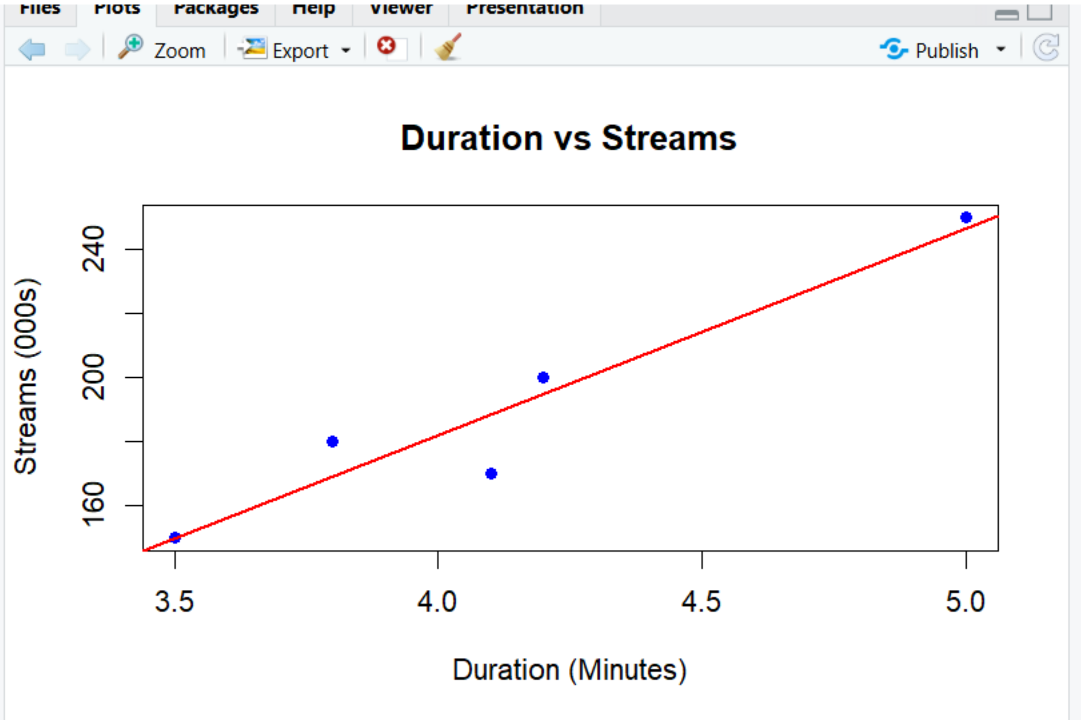


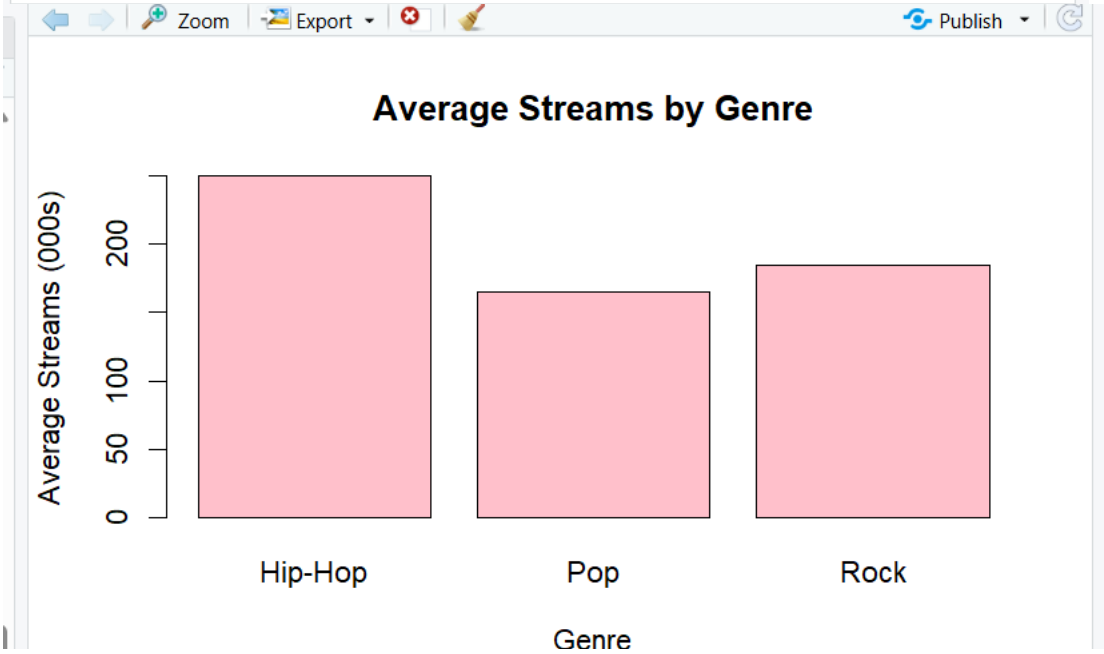


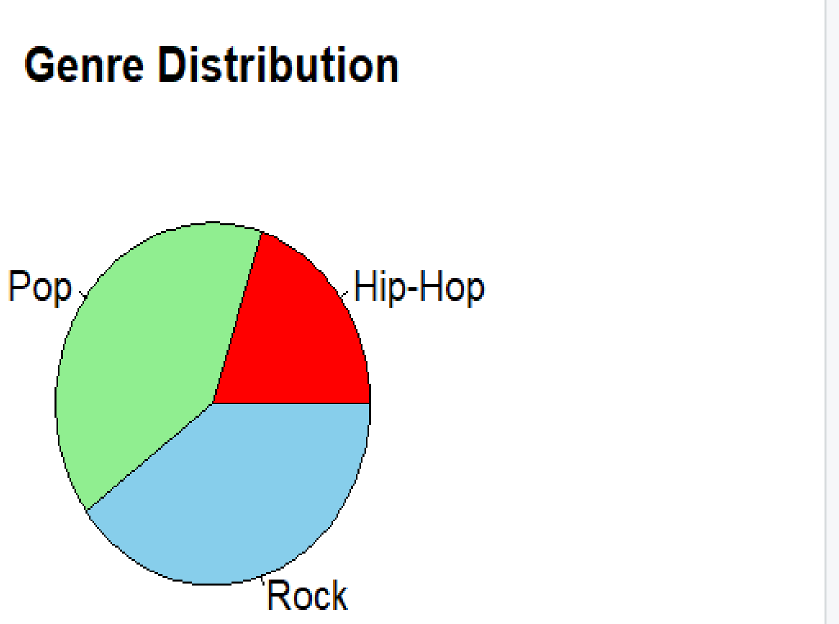
30.Code:

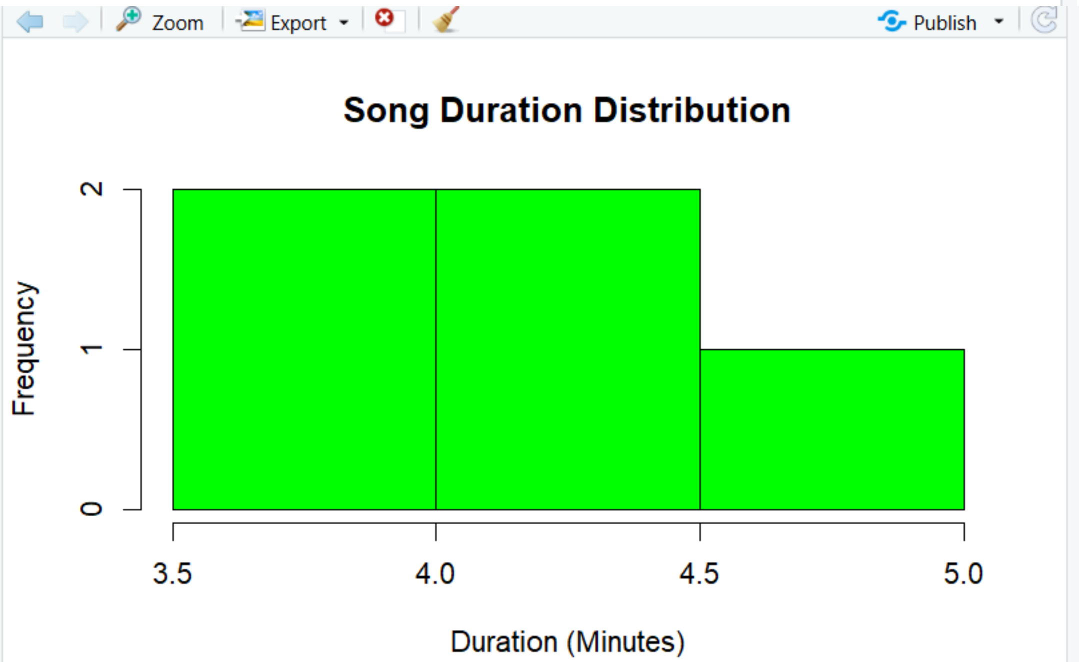
```
Song_ID <- c(1,2,3,4,5)
Duration <- c(3.5,4.2,3.8,5.0,4.1)
Streams <- c(150,200,180,250,170)
Genre <- c("Pop","Rock","Pop","Hip-Hop","Rock")
music <- data.frame(
  Song_ID,
  Duration,
  Streams,
  Genre
)
print(music)
```

```
hist(music$Duration,
     main="Song Duration Distribution",
     xlab="Duration (Minutes)",
     ylab="Frequency",
     col="green",
     border="black")
genre_table <- table(music$Genre)
pie(genre_table,
    main="Genre Distribution",
    labels=names(genre_table),
    col=c("red","lightgreen","skyblue"))
avg_streams <- aggregate(Streams ~ Genre,
                         data=music,
                         mean)
barplot(avg_streams$Streams,
        names.arg=avg_streams$Genre,
        main="Average Streams by Genre",
        xlab="Genre",
        ylab="Average Streams (000s)",
        col="pink",
        border="black")
plot(music$Duration,
     music$Streams,
     main="Duration vs Streams",
     xlab="Duration (Minutes)",
     ylab="Streams (000s)",
     pch=19,
     col="blue")
abline(lm(Streams ~ Duration,
          data=music),
       col="red",
       lwd=2)
```


Files
Plots
Packages
Help
Viewer
Presentation
Zoom
Export
Publish
Duration vs Streams
Streams (000s)
240
200
160
3.5
4.0
4.5
5.0
Duration (Minutes)


Zoom
Export
Publish
Average Streams by Genre
Average Streams (000s)
200
100
50
0
Hip-Hop
Pop
Rock
Genre


Genre Distribution
Pop
Hip-Hop
Rock


Zoom
Export
Publish
Song Duration Distribution
Frequency
0
1
2
3.5
4.0
4.5
5.0
Duration (Minutes)